UNIT 4 · TOPIC 1(A)

Thermodynamics and Air-Standard Power Cycles

Syllabus topic: Engineering thermodynamics, power cycles and fuels

Systems, properties and processes

- **Systems:** closed (mass fixed — gas in the cylinder), open (mass crosses the boundary — turbine, radiator), isolated.
- **Properties:** intensive (p, T) vs extensive (V, U, H).

$$\text{Ideal gas: } pV = mRT;\ R_{air} = 0.287\ \text{kJ/kg·K};\ c_p - c_v = R;\ \gamma = c_p/c_v = 1.4 \text{ for air } (c_p = 1.005,\ c_v = 0.718\ \text{kJ/kg·K})$$

Process	Condition	Work / use
Isochoric	V constant	W = 0
Isobaric	p constant	W = pΔV
Isothermal	T constant	pV = constant
Adiabatic / isentropic	No heat transfer	pV^γ = constant — ideal engine compression and expansion
Polytropic	pV^n = constant	Real processes, n ≈ 1.3

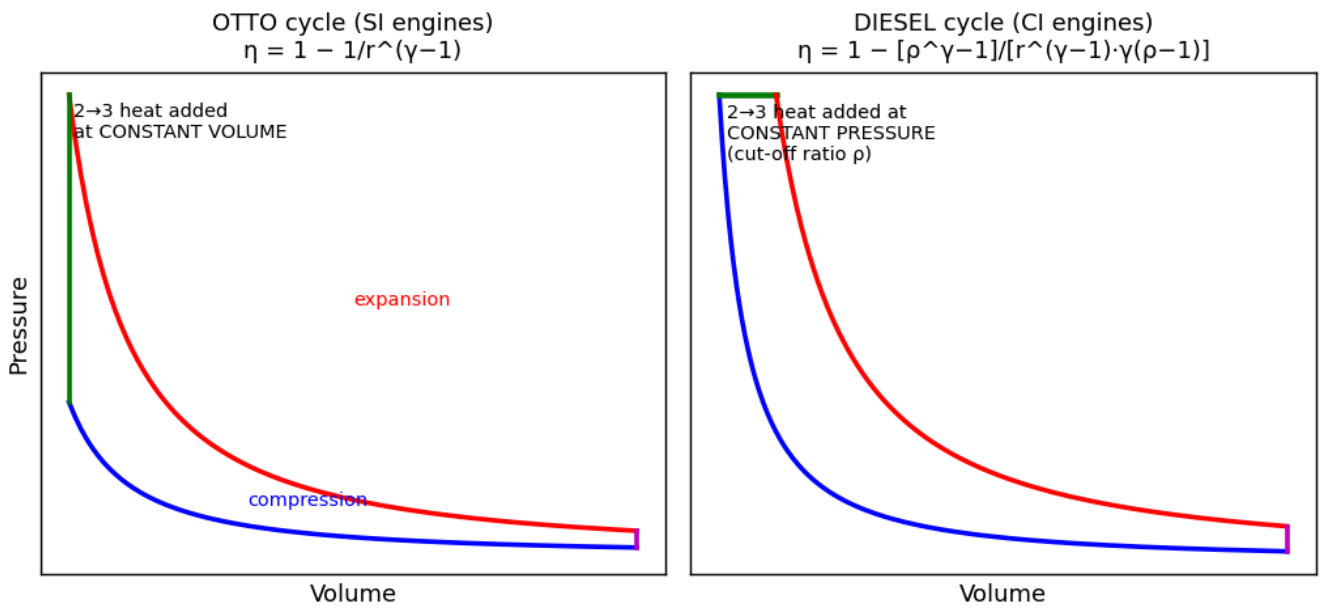
Laws of thermodynamics

- **First law** — energy is conserved: Q = ΔU + W. Over a cycle, net heat = net work.
- **Second law** — heat cannot be converted fully and continuously into work (Kelvin–Planck), and heat cannot flow unaided from cold to hot (Clausius). It defines entropy and limits efficiency.

$$\text{Carnot (the ceiling): } \eta_{carnot} = 1 - T_{low}/T_{high} \text{ (absolute temperatures)}$$

Example. Source 900 K, sink 300 K → η = 1 - 300/900 = **66.7%**. No real engine beats this; real tractor diesels reach ~**35–42%** brake thermal efficiency.

Air-standard power cycles



Otto and Diesel cycles on the p–V diagram

Otto cycle (SI / petrol / gas engines)

1. Isentropic compression.
2. Heat added at **CONSTANT VOLUME** (spark).
3. Isentropic expansion.
4. Heat rejected at constant volume.

$$\eta_otto = 1 - 1/r^{(\gamma-1)}$$

Example. r = 8, γ = 1.4: 8^0.4 = 2.297 → η = 1 - 0.435 = **56.5%** (air-standard).

KNOCKING (auto-ignition of the end gas) limits the compression ratio to ~**6–10** — hence the octane requirement.

Diesel cycle (CI engines)

1. Isentropic compression.
2. Heat added at **CONSTANT PRESSURE** (fuel injected into hot air).
3. Isentropic expansion.
4. Heat rejected at constant volume.

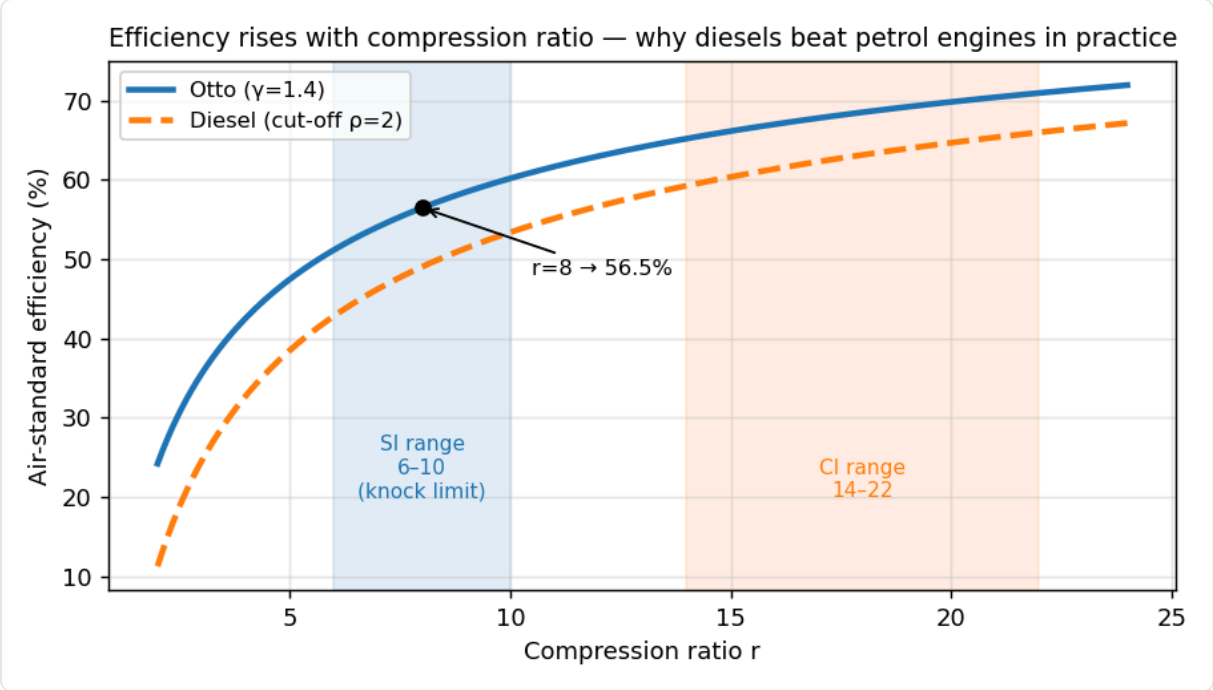
$$\eta_diesel = 1 - [\rho^{\gamma} - 1] \,/\, [r^{(\gamma-1)} \cdot \gamma \cdot (\rho - 1)],\ \rho = \text{cut-off ratio} = V_3/V_2$$

- Efficiency FALLS as ρ (load) rises.
- r = **14–22** is practicable because only AIR is compressed — the fuel sets no knock limit.

Dual (mixed) cycle and comparisons

- **Dual / limited-pressure cycle:** heat added partly at constant volume, partly at constant pressure — closest to modern high-speed diesels.

- **Same compression ratio:** η_otto > η_dual > η_diesel.
- **Same maximum pressure/temperature:** the Diesel cycle is the most efficient.
- In practice diesels win because their r is far higher (figure below).



Air-standard efficiency rises with compression ratio — SI range 6–10, CI range 14–22

Mean effective pressure (MEP) = net cycle work ÷ swept volume

MEP is a size-independent measure of specific output and the basis of the indicated-power formula (Unit 3, Topic 12a; Topic 3a here).

Practice MCQs

Q1. For air, the ratio of specific heats γ is:

(a) 1.2 (b) 1.3
(c) 1.4 (d) 1.67

Q2. cp - cv equals:

(a) γ (b) R
(c) U (d) H

Q3. In an isochoric process, work done is:

(a) pΔV (b) Zero
(c) Maximum (d) Negative always

Q4. Ideal engine compression/expansion is modelled as:

(a) Isothermal (b) Isobaric
(c) Isentropic (pV^γ = const) (d) Isochoric

Q5. The first law of thermodynamics states:

(a) Entropy rises (b) Q = ΔU + W (energy conservation)
(c) Heat flows cold→hot (d) Work is heat

Q6. Maximum possible efficiency between 900 K and 300 K is:

(a) 33.3% (b) 50%

(c) 66.7% (d) 100%

Q7. Heat addition in the Otto cycle occurs at constant:

(a) Pressure (b) Volume

(c) Temperature (d) Entropy

Q8. Heat addition in the Diesel cycle occurs at constant:

(a) Volume (b) Pressure

(c) Entropy (d) Enthalpy

Q9. Air-standard Otto efficiency is:

(a) 1 - r^(γ-1) (b) 1 - 1/r^(γ-1)

(c) 1/r^γ (d) 1 - T2/T1 always

Q10. For r = 8 and γ = 1.4, Otto efficiency ≈

(a) 46.5% (b) 51%

(c) 56.5% (d) 62%

Q11. Raising the cut-off ratio ρ (higher load) makes Diesel-cycle efficiency:

(a) Rise (b) Fall

(c) Constant (d) Double

Q12. For the SAME compression ratio, the most efficient cycle is:

(a) Diesel (b) Dual

(c) Otto (d) All equal

Q13. For the same MAXIMUM pressure and temperature, the most efficient is:

(a) Otto (b) Diesel

(c) Dual (d) Equal

Q14. Modern high-speed diesels are best represented by the:

(a) Otto cycle (b) Dual (limited-pressure) cycle

(c) Carnot (d) Brayton

Q15. SI-engine compression ratio is limited by:

(a) Cost (b) Knocking of end gas

(c) Weight (d) Cetane

Q16. Typical CI (diesel) compression ratios are:

(a) 4–6 (b) 6–10

(c) 14–22 (d) 25–35

Q17. Diesels can use high r because:

(a) Fuel is compressed (b) Only AIR is compressed — fuel injected later

(c) They run cold (d) They are heavy

Q18. Mean effective pressure equals:

(a) Peak pressure (b) Net cycle work ÷ swept volume

(c) Injection pressure (d) Boost

Answers

Q1 — (c) 1.4

Q2 — (b) R

Q3 — (b) Zero

Q4 — (c) Isentropic (pV^γ = const)

Q5 — (b) Q = ΔU + W (energy conservation)
Q6 — (c) 66.7% — *η = 1 - 300/900 = 2/3 = 66.7%.*
Q7 — (b) Volume
Q8 — (b) Pressure
Q9 — (b) 1 - 1/r^(γ-1)
Q10 — (c) 56.5% — *8^0.4 = 2.297 → 1 - 0.435 = 56.5%.*
Q11 — (b) Fall
Q12 — (c) Otto — *At equal r, constant-volume (Otto) heat addition is thermodynamically superior.*
Q13 — (b) Diesel — *At equal peak pressure, the Diesel cycle rejects less heat — the classic viva pair with Q12.*
Q14 — (b) Dual (limited-pressure) cycle
Q15 — (b) Knocking of end gas
Q16 — (c) 14–22
Q17 — (b) Only AIR is compressed — fuel injected later
Q18 — (b) Net cycle work ÷ swept volume

Sources: Liljedahl, Turnquist, Smith & Hoki — Tractors and their Power Units; standard thermodynamics texts; PIB/MoPNG releases on ethanol blending; MoRTH/CMVR TREM notifications and DieselNet (as verified in the source notes, July 2026); NPTEL lectures.